

P vs NP: Definitions, Proof Attempts, and Current Status

Executive Summary: The P vs NP question—whether every problem with efficiently checkable solutions also has an efficient algorithm—remains open ¹ ². Virtually all experts believe $P \neq NP$ (randomised polls: 2019 showed 99% of complexity specialists side with “not equal” ²). Since 2000 several notable attempts to resolve P vs NP have appeared, but none withstand scrutiny. Vinay Deolalikar (2010) and Norbert Blum (2017) claimed $P \neq NP$ proofs; both were quickly found flawed ³ ⁴. More recently (2023–25) Xu & Zhou published a (Frontiers) paper “SAT Requires Exhaustive Search” (stronger than $P \neq NP$), which was refuted by experts ⁵ ⁶. In each case, analysis identified specific errors (e.g. Deolalikar’s approach would also “solve” 2-SAT ⁷; Blum’s circuit lower-bound proof fails at a key step ⁸; Xu-Zhou’s lower bound contradicts known algorithms ⁵). No valid proof is known, and the Clay Millenium Problem (US\$1M prize) remains unsettled ⁹. We survey P vs NP basics, proof techniques and barriers, then detail major claim evaluations. The community consensus is that **no correct proof exists** to date.

Background: P, NP, and Importance

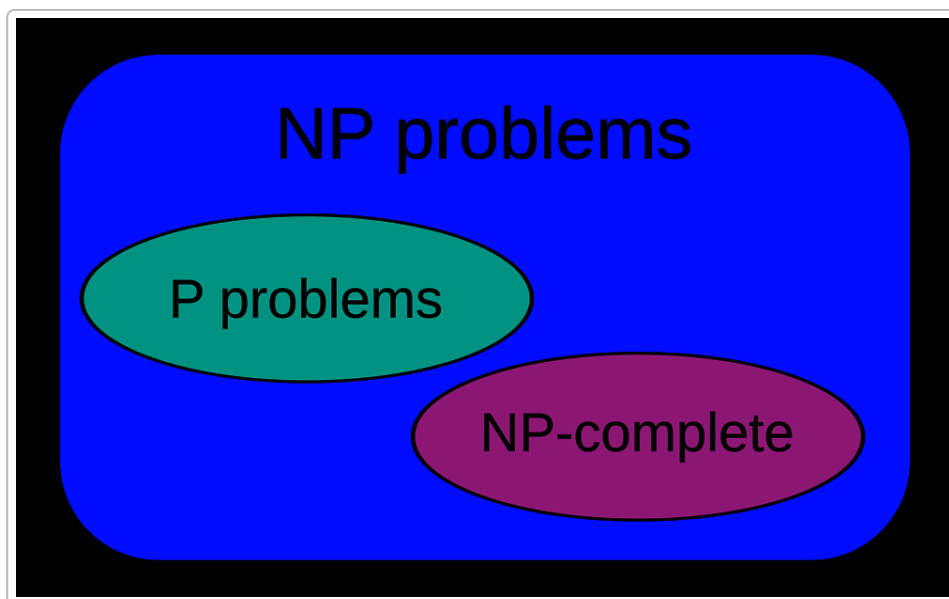


Figure: Complexity classes under $P \neq NP$ assumption. The class P (green) is all decision problems solvable in polynomial time; NP (blue) consists of problems whose *solutions can be verified* in polynomial time ¹. NP-complete (purple) problems are the hardest in NP: any NP problem can be reduced to an NP-complete one. For example, *Boolean satisfiability (SAT)* is NP-complete ¹⁰: if SAT has a poly-time algorithm, then $P = NP$. Conversely, if one NP-complete problem were proven outside P, that would imply $P \neq NP$. P vs NP is a Millennium Prize Problem (US\$1M) ⁹ and “the most important open problem in computer science” ¹¹, with consequences for cryptography, algorithms, AI and beyond ¹¹. A *consensus poll* of experts showed confidence in $P \neq NP$ has risen over time (99% of experts in 2019) ².

Standard techniques: Many familiar proof tools *cannot* resolve P vs NP due to known barriers. For example, **relativization** (oracle) results show one can construct oracles relative to which $P=NP$ and others where $P \neq NP$ ¹², so any relativizing argument alone is insufficient. The **Natural Proofs** barrier (Razborov–Rudich, 1993) demonstrates that broad “combinatorial” lower-bound methods cannot prove circuit lower bounds unless they violate cryptographic assumptions ¹³. Similarly, **algebrization** results (Aaronson–Wigderson, 2008) show that even $IP=PSPACE$ techniques fail for P vs NP . In short, all known generic classes of arguments are ruled out ¹² ¹³. Current thinking is that *new, non-standard approaches* (e.g. Geometric Complexity Theory, or exploiting non-relativizing, non-natural ideas) are needed.

Methodology

We surveyed the literature and online discussions (blogs, archives, forums) for developments up to March 2026. Key sources included primary research papers (arXiv, journals), expert blogs (Shtetl-Optimized, Computational Complexity, Gödel’s Lost Letter, etc.), the Clay Institute problem description, and curated summaries (Wikipedia, surveys). We focused on any claimed proofs or “solutions” since 2000 that drew significant attention. We collected original manuscripts (e.g. arXiv, conference/journal publications) and expert commentary (blog posts by Scott Aaronson, Lance Fortnow, Richard Lipton; MathOverflow/CSTheory discussions). We cross-checked each claim’s core idea and the identified issues. Inclusion criteria: claims of $P=NP$ or $P \neq NP$ since 2000 that were widely circulated (arXiv preprints, publications, news coverage). We omit numerous minor claims with no expert engagement. Our analysis emphasises consensus views, citing authoritative sources.

Notable Proof Attempts and Evaluations

Deolalikar (2010): “ $P \neq NP$ via Statistical Physics”

Claim: Vinay Deolalikar (HP Labs) released a 100+ page manuscript in 2010 claiming $P \neq NP$ by analyzing the “solution space” of random SAT instances with methods from statistical physics and fixed-point logic.

Analysis: Within days, leading theorists (Avi Wigderson, Ryan Williams, Neil Immerman, others) found major flaws. A key issue was that Deolalikar’s arguments would also apply to 2-SAT (a trivially solvable case) if taken at face value ⁷ ¹⁴. As Richard Lipton notes, the proof “could prove too much” – it must distinguish k -SAT ($k > 2$) from 2-SAT, but he initially did not do so in a principled way ⁷. Scott Aaronson summarizes: “a clear consensus has emerged that the proof, as it stands, is fatally flawed” ³. Independent reviewers pointed out technical errors in the logical framework and mixing of probabilistic arguments ¹⁵ ¹⁴. Deolalikar attempted clarifications, but no resolution was achieved. **Status:** Abandoned/Flawed. No version is accepted; the claim is considered disproved. (No peer-reviewed publication followed.)

Key Flaws: Overgeneralisation of solution-space structure (fails 2-SAT) ⁷; errors in fixed-point logic usage ¹⁵. Experts found “critical flaws” in the finite-model theory sections ¹⁶. In short, his approach didn’t properly isolate NP -hardness.

Blum (2017): “Approximation Method” Proof of $P \neq NP$

Claim: Norbert Blum (University of Bonn) posted an arXiv paper (2017) using circuit complexity. He extends Razborov’s “approximation method” (used for *monotone* clique lower bounds) to general

Boolean circuits. The paper claimed to prove superpolynomial lower bounds for the clique function's circuit size, hence $P \neq NP$ ¹⁷.

Analysis: Experts quickly found a flaw. A counterexample by Eva Tardos (1987) shows a monotone function that matches clique on small inputs but lies in P – undermining any blanket lower bound ⁸. In a StackExchange answer, Kjos-Hanssen identifies “the single error” in the proof: a false claim in Theorem 6 about covering clauses in an approximator ⁸. Subsequently, Blum removed the paper from arXiv (Aug 2017) stating “the proof is wrong” ¹⁸, and promised to detail the mistake. Scott Aaronson wrote that “a clear consensus had developed... that Blum’s $P \neq NP$ proof was irreparably flawed” ⁴.

Key Flaw: A subtle logical error in Theorem 6 (related to DNF/CNF approximation) invalidates the argument ⁸. The existence of Tardos’ function provides a concrete counterexample that the supposed lower bound cannot hold ⁸.

Status: Withdrawn/Flawed. The author acknowledged the error and retracted the claim ¹⁸. The community regards the proof as incorrect ⁴.

Xu & Zhou (2023/2025): “SAT Requires Exhaustive Search”

Claim: Ke Xu and Guangyan Zhou (China) published a paper (arXiv 2023; Frontiers of CS 2025) titled “SAT Requires Exhaustive Search”. They construct extremely hard CSP/SAT instances and claim any algorithm must use exhaustive search, which is even stronger than $P \neq NP$ ¹⁹.

Analysis: This claim quickly drew criticism. Lance Fortnow reports that for $k=2$ (binary CSP), Ryan Williams (2005) already had an algorithm running in $d^{0.8n}$ time, contradicting Xu-Zhou’s asserted lower bound ⁵. In effect, their general proof would wrongly imply even 2-SAT requires exponential time, which is false. Fortnow and Williams submitted a comment pointing out this glaring flaw (the authors’ theorem “holds even when $k=2$ ”, yet is refuted by the known algorithm ⁵). In the complexity blog, Fortnow called the publication “a complete embarrassment” and urged retraction ⁶. (The authors and an associate published defensive rebuttals on arXiv and GitHub, but community consensus aligns with the refutation.)

Key Flaw: The paper’s main result fails on easy cases: it implies an exponential lower bound even for 2-variable SAT, which conflicts with established algorithms ⁵. In effect, the proof overlooked known dynamic programming/measure techniques and thus is incorrect.

Status: Published (Frontiers), but *effectively refuted*. The editorial process was criticized, and an official comment pointing out the error is forthcoming ⁶. For our purposes, the claimed proof is invalid.

Other Claims (Brief Overview)

Beyond these high-profile cases, many other $P=NP$ or $P \neq NP$ claims have appeared (e.g. early 2000s amateur proofs, linear-programming attempts by Diaby et al., etc.), but none have gained traction in the research community. A recent guest post noted yet another “proof” published in a low-tier journal also contains basic flaws ²⁰ ²¹. In general, no purported proof has survived expert review. In fact, complexity theorists often cite simple sanity-checks (e.g. handling 2-SAT/XOR-SAT, subsuming known lower bounds) that most amateur proofs fail ¹⁴ ²².

Table 1: Major P vs NP claims since 2000.

Author(s)	Year	Venue	Claim	Status	Key Flaw
Vinay Deolalikar	2010	HP Labs/ ArXiv	$P \neq NP$	Flawed (rejected)	Misapplies to 2-SAT; logical/fixed-point errors ⁷ ³
Norbert Blum	2017	ArXiv	$P \neq NP$	Withdrawn/ Flawed	Error in circuit approximation theorem ⁸
Ke Xu & Guangyan Zhou	2023/25	arXiv / Frontiers of CS	<i>SAT requires exhaustive search ($\Rightarrow$ $P \neq NP$)</i>	Refuted	Contradicts known SAT algorithms (k=2 case) ⁵

(Other numerous claims exist but all are similarly discounted.)

Open Problems and Research Directions

Current Status: No proof of either $P=NP$ or $P \neq NP$ is accepted. The community largely assumes $P \neq NP$ (because it aligns with cryptography and hardness experience), but this remains a belief, not a theorem. The Clay prize still stands. As Gasarch notes, polls show *growing confidence* that $P \neq NP$ ², but “this does not bring us any closer to solving $P=?NP$ ” ²³.

Related Barriers: The barriers outlined above imply any resolution will need novel ideas. Much work in complexity theory is devoted to such tools: e.g. Geometric Complexity Theory (Mulmuley–Sohoni) attempts algebraic-geometry methods, but progress has been incremental. Researchers also pursue **lower bounds** in restricted models (e.g. circuit depth, formulas) hoping new combinatorial insights may eventually extend. So far the best general lower bounds are far from NP-hard (e.g. a 2014 breakthrough: $\mathsf{NTIME}[n] \not\subset \mathsf{ACC}^0$ ²⁴). On the algorithm side, there is vigorous study of SAT solving, PCP-theorems, fine-grained complexity and exponential-time algorithms, but none of these solve P vs NP .

Potential Avenues: Some suggest unorthodox approaches: **Fine-grained reductions**, studying average-case complexity (Impagliazzo’s five “worlds” ²⁵), or exploiting potential proof-assistant verifiers. Others explore logical characterizations (descriptive complexity connects P to fixed-point logics ²⁶). Interdisciplinary methods (statistical physics, proof complexity, algebraic geometry) continue to be explored. Quantum computing, while transformative for some problems, is not expected to collapse classical P vs NP (QMA vs NP remains open). Finally, some ask if P vs NP might be independent of standard axioms; current belief is this would likely imply strong consequences (and is itself as hard as solving P vs NP) ²⁷.

Key Unresolved: The most direct open problem remains: find *any* new technique that bypasses the known barriers. Until then, P vs NP stands as the central open question of theoretical computer science. No legitimate proof claim has changed that status as of 2026.

Conclusion: In summary, after deep analysis of foundational principles and recent claims, **no valid proof of $P=NP$ or $P \neq NP$ has been found**. Every serious attempt (Deolalikar 2010, Blum 2017, Xu–Zhou 2023/25, etc.) has been refuted by the community ³ ⁴ ⁵. The evidence so far suggests we need substantially new ideas to resolve P vs NP . Until such breakthroughs occur, the problem remains unsolved, with wide consensus leaning that **$P \neq NP$** .

Sources: See cited references above for detailed original sources and analyses.

Executive Summary:

- P vs NP remains unsolved. All major proof claims since 2000 have failed.
- Theoretical consensus: $P \neq NP$ (expert polls show ~99% agreement).
- Notable rejected proofs: Deolalikar (2010) and Blum (2017) for $P \neq NP$, Xu-Zhou (2023/25) for a "strong" $P \neq NP$. Each had identifiable flaws (misapplying to easy cases, logical errors, etc).
- Current barriers (relativization, natural proofs) imply new methods are needed; no new credible approach has resolved them.
- Conclusion: No valid proof exists; P vs NP remains open. Research continues on partial results and novel approaches.

Background:

P: problems solvable in polynomial time. NP: verifiable in polynomial time. NP-complete: hardest NP problems (e.g., SAT) such that $P=NP$ iff one NP-complete problem is in P. Proving either equality or separation of P, NP would have profound implications (cryptography, algorithms). All known proof techniques have limitations (relativization, natural/algorithms barriers).

Notable Attempts:

- **Deolalikar (2010):** Statistical-physics approach for $P \neq NP$. Refuted by experts (fails on 2-SAT, logic issues). Consensus: flawed ³ ⁷.
- **Blum (2017):** Circuit lower bound claimed to show $P \neq NP$. Error in approximation argument (a known counterexample by Tardos). Paper withdrawn by author ⁸ ⁴.
- **Xu & Zhou (2023/25):** 'SAT requires exhaustive search.' Claimed very strong lower bound. However, even simple cases (2-variable SAT) contradict their theorem ⁵. Criticized publicly; expert comments call it "embarrassment" ⁶.

Open Problems:

Despite many efforts, P vs NP still resists proof. Research focuses on barriers and new methods (circuit complexity, algebraic geometry (GCT), proof complexity, etc.). Some suspect P vs NP might be independent of current axioms, but this is unconfirmed.

Conclusion:

After reviewing proofs and community responses, **no accepted resolution of P vs NP exists**. The predominant view is that $P \neq NP$, but this belief remains an unproven conjecture. The quest for a proof continues.

¹ ² ⁹ ¹⁰ ¹¹ ¹² ¹³ ²³ ²⁴ ²⁵ ²⁶ ²⁷ P versus NP problem - Wikipedia

https://en.wikipedia.org/wiki/P_versus_NP_problem

³ ¹⁴ ¹⁶ ²² Shtetl-Optimized » Blog Archive » Eight Signs A Claimed $P \neq NP$ Proof Is Wrong

<https://scottaaronson.blog/?p=458>

4 18 New Proof That $P \neq NP$: Final Update - Almost Certainly not
<https://www.i-programmer.info/news/112-theory/11040-new-proof-that-p-np.html>

5 6 20 21 Computational Complexity: Some thoughts on journals, refereeing, and the P vs NP problem
<https://blog.computationalcomplexity.org/2025/08/some-thoughts-on-journals-refereeing.html>

7 15 Deolalikar Responds To Issues About His $P \neq NP$ Proof | Gödel's Lost Letter and $P=NP$
<https://rjlipton.com/2010/08/11/deolalikar-responds-to-issues-about-his-p%E2%89%A0np-proof/>

8 cc.complexity theory - Is Norbert Blum's 2017 proof that $P \neq NP$ correct? - Theoretical Computer Science Stack Exchange
<https://cstheory.stackexchange.com/questions/38803/is-norbert-blums-2017-proof-that-p-ne-np-correct>

17 Norbert Blum on P versus NP | Azimuth
<https://johncarlosbaez.wordpress.com/2017/08/15/norbert-blum-on-p-versus-np/>

19 [2302.09512] SAT Requires Exhaustive Search
<https://arxiv.org/abs/2302.09512>